

Alex Gates — Viable Peer Congestion Model (interpreted)

Structured notes with LaTeX. Verbatim source: `Alex_model.md`.
Live discussion addendum: [Paper Directions 12 \(PD12\)](#) — May 20, 2026 transcript; clarifies capability, crowding vs. quality, and empirical priorities.

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Main idea

Systems that concentrate high-performing individuals generate **endogenous congestion** in evaluative comparisons, producing **diminishing and eventually negative marginal returns** to elite affiliation.

Congestion is **not** “more good people nearby.” It must alter how **evaluation** and **organizational selection** operate.

Core simulation (proof of concept)

Latent ability → assortative teams → **viability** (smooth threshold) → **peer congestion** (density of viable peers, LOO) → **evaluation score** → global selection (top decile).

Key objects:

Symbol	Meaning
a_i	Own ability
$\bar{a}_t$	Team mean ability
v_i	Individual viability, e.g. $\sigma(\gamma(a_i - \theta_v))$
$C_{i,t}$	Peer congestion (mean viable-peer density on team, excluding self)
S_i	Evaluation score: $a_i - \lambda C_{i,t} + \varepsilon_i$

Implementation: `sports/539_alex_model.ipynb` — function `simulate_viable_peer_congestion()`.

Mechanisms (not mutually exclusive)

1. Comparative evaluation

Evaluators compare candidates to **local reference groups**. In elite environments, “good” becomes normal; **exceptional performances appear less stand-out** (PD12 wording: not “less exceptional,” but harder to distinguish from strong peers).

$$\text{signal}_i = a_i - f(\text{local peer quality}) + \epsilon_i$$

—not a_i alone. (NCAA: 15 PPG on a mid-tier team vs. the same player on a stacked roster.)

Prediction: congestion affects **signal distinctiveness**.

2. Attribution dilution

On strong teams, success is harder to assign individually (system? teammates? suppressed stats?). Plausible in academia, sports, firms, entertainment.

3. Finite attention

Bounded evaluator capacity. Elite teams contain **too many** plausible candidates → triage on stars/heuristics → harms **marginally excellent** candidates.

Prediction: congestion penalty strongest for **near-threshold** individuals (superstars survive; weak never competed; borderline elite are substitutable).

4. Organizational risk minimization

Many viable candidates → conservative selection; marginal benefit of any one hire declines; reliance on prototypes, fame, age heuristics. **Substitutability** rises.

5. Opportunity suppression (recursive)

Elite teams allocate minutes, touches, visibility, recommendations unevenly:

$$\text{viable peers} \rightarrow \text{fewer opportunities} \rightarrow \text{less distinct signals} \rightarrow \text{more congestion.}$$

6. Queueing and timing

Few advancement slots; many qualified people queue → delay, exit, invisibility — **temporal** congestion.

Framing

Elite environments increase both **capability** and **substitutability**.

Environment	Capability	Substitutability
Weak	Low	Low
Moderate	High	Manageable
Elite	Very high	Extreme

PD12 clarification on “capability”: In the May 20 conversation this was **not** organizational/program capability (e.g. “Duke’s institutional strength”). Alex revised it to **own performance and differentiation within the local peer group** — effectively a **proxy for latent talent** a_i : in moderate environments top players still **stand out**; in elite environments everyone is strong but **substitutable**, so identifiability collapses even when absolute a_i is high. See the **PD12 addendum** below for full discussion.

Organizations reward quality imperfectly under uncertainty, finite attention, and **comparative** evaluation. Many highly viable candidates → compressed margins, noisier/heuristic selection.

NCAA basketball operationalization

Observable **viable peer** structure. Three constructs:

1. **Own ability** a_i — e.g. points per minute; better: NBA predictive composite (BPM, WS/40, recruiting, etc.). PPM acceptable for proof of concept.
2. **Team quality**

$$\bar{a}_t = \frac{1}{n_t} \sum_j a_j.$$

3. **Viable-peer congestion** — density of teammates who **compete for the same scarce advancement**, not merely “good teammates.”

Simple congestion:

$$C_{i,t} = \frac{1}{n_t - 1} \sum_{j \neq i} \mathbf{1}(a_j > \theta),$$

with θ from top prospect share, draft probability, McDonald’s All-American, future pro, combine invite, etc.

Role-specific (stronger):

$$C_{i,t}^{\text{role}} = \frac{1}{n_t - 1} \sum_{j \neq i} w_{ij} \mathbf{1}(a_j > \theta),$$

w_{ij} = positional / playstyle similarity.

Empirical draft model:

$$P(\text{draft}_i) = f(a_i, \bar{a}_t, C_{i,t}).$$

Predictions: $\partial a_i > 0$; $\bar{a}_t$ initially positive; $\partial C_{i,t} < 0$.

Inverted U: raw team quality may show downturn; **viable-peer congestion** should explain it.

- Baseline: $\text{Draft}_i \sim a_i + \bar{a}_t + \bar{a}_t^2$
- Mechanism: $\text{Draft}_i \sim a_i + \bar{a}_t + C_{i,t}$ — quadratic weakens

Heterogeneity: downturn strongest for **middle-high** prospects (not top 1%, not non-prospects).

Quasi-experiments: mega-recruit years, portal shocks, one-and-done, NIL concentration — e.g. after several elite guards arrive, incumbent guards’ draft odds fall **conditional on performance**.

Secondary empirical tests (beyond curve-fitting)

Core test: downturn **mediated** by viable-peer density.

1. Mediation

Baseline:

$$Y_i \sim a_i + \bar{a}_t + \bar{a}_t^2.$$

Mechanism:

$$Y_i \sim a_i + \bar{a}_t + \bar{a}_t^2 + C_{i,t}, \quad C_{i,t} = \frac{1}{n_t - 1} \sum_{j \neq i} \mathbf{1}(a_j > \theta)$$

(or smooth viability version). Expect $C_{i,t} < 0$ and **shrinkage** of the negative quadratic / top-end team-quality penalty.

2. Congestion concentration

Bin teams by $C_{i,t}$. High-congestion bins → steepest right-side decline; low-congestion → monotone or saturating returns to $\bar{a}_t$.

3. Assortativity interaction

$$Y_i \sim a_i + \bar{a}_t + \bar{a}_t^2 + A_s \cdot \bar{a}_t^2,$$

A_s = system assortativity. Negative interaction: stronger sorting → stronger top-end penalty.

Further predictions

Prediction	Test sketch
Mediation	Add $C_{i,t}$; quadratic on $\bar{a}_t$ weakens
Near-threshold harm	$Y_i \sim a_i + \bar{a}_t + C_{i,t} + a_i \times C_{i,t}$; penalty largest just above viability margin
External benchmarks	Penalty smaller for standardized metrics; larger for committees / prestige narratives
Fixed opportunity capacity	Penalty stronger when few slots per team/lab
Delay, not only failure	Survival $h_i(t) = h_0(t) \exp(\beta_1 a_i + \beta_2 \bar{a}_t + \beta_3 C_{i,t})$, $\beta_3 < 0$ for high a_i
Cohort shock	DiD: existing near-threshold members crowded when strong cohort enters
Mobility reversal	Leaving congested elite for strong-but-less-congested teams raises success odds

Paper framing:

Beyond reproducing an inverted-U between team quality and individual advancement, the model predicts the downturn is explained by **viable-peer density**, concentrated among **near-threshold** candidates, amplified in **assortative** systems, and partially **reversed** by moves to less congested environments.

Link to this repo (538 / 530)

Alex construct	Current PDE stack
a_i	<code>perf</code> / synthetic <code>ability</code>
$\bar{a}_t$	<code>team_mean</code> / pool mean
Viable peer $C_{i,t}$	Not raw LOO sum of all A_j ; use $\sum_{j \neq i} \mathbf{1}(a_j > \theta)$ or smooth viability density
LOO mean L_Q	<code>poolq_loo</code> / <code>congestion_quality</code>
Raw LOO sum L_C	<code>pool_c_loo</code> — scales with roster size; near redundant with L_Q at fixed n

Next modeling step: replace or supplement `pool_c_loo` with **viable-peer congestion** as in the simulation notebook.

Addendum: Paper Directions 12 (PD12) — May 20, 2026

Source: `transcripts/20260520_Paper_Directions_12_otter_ai-transcript.docx` (Otter; ~16 min).

Participants: Alex Gates, Charles Levine.

Purpose of this section: Record what was **agreed, corrected, and prioritized** in the live discussion so implementation and writing stay aligned with Alex’s intent—not only with the AI-generated text in `Alex_model.md`.

Provenance of `Alex_model.md`

Near the start of PD12, Charles noted that Alex had not literally written every sentence of the memo by hand. Alex confirmed the document was produced in **~45 minutes with AI assistance** from prompts (“we have the tools, let’s use them”). Treat `Alex_model.md` as a **structured theoretical brief** and mechanism checklist, not a finalized paper. **PD12 overrides or sharpens** several terms (especially **capability** and **crowding vs. quality**) where the conversation went deeper than the draft text.

Mechanism-by-mechanism notes from PD12

The six mechanism classes in `Alex_model.md` were walked through in order. Below: what resonated, what to stress empirically, and what **not** to over-build in v1.

1. Comparative evaluation

- Charles reframed the second bullet: the punchline is that **exceptional performances appear less stand-out** in elite reference groups—not that they become “less exceptional” in absolute terms.
- Alex endorsed that wording.
- **Empirical tie-in:** This is **signal distinctiveness** (evaluator sees $a_i - f(\text{local peers})$), not raw production. Pool **mean** quality ($\bar{a}_t$, `poolq_100`) is related but not sufficient for congestion.

2. Attribution dilution

- Uncontroversial; aligns with prior VECTOR thinking (lab effect, system player, brand halo).
- No PD12 change; include in narrative, optional as separate test later.

3. Finite attention

- Strong alignment with Charles’s **earlier model** (some variables already mirrored this)—good convergence check.
- **Substitutability** of **borderline elite** candidates is the money phrase.
- Charles linked borderline substitutables to the **NIH / Wang** “just at the cut line” population: the inverted-U harm is not for obvious superstars or non-contenders, but for **near-threshold** types who look interchangeable when many similar peers exist.

4. Organizational risk minimization

- Charles: “that’s **Moneyball**.”
- NBA gloss: when several teammates look promising, scouts fall back on **heuristics** (physically prototypical, younger, already famous)—Charles jokingly added “or white” as a cynical example of voodoo selection; keep that out of formal write-ups unless explicitly studying bias.
- **Modeling:** This supports **noisier, less meritocratic selection** when $C_{i,t}$ is high, not only lower $P(\text{draft})$.

5. Opportunity suppression (recursive)

- Charles strongly endorsed Alex’s word “**recursive**”: viable peers → fewer touches/minutes/visibility → weaker observable stats → even less distinction → more congestion.
- This bridges **production** and **evaluation**; important for NCAA where a_i is endogenous to role allocation.

6. Queueing and timing

- Charles asked if this is just **selection scarcity**; Alex: **yes**, but scarcity can arise in **multiple ways**—time (queue), comparison (local rank), finite slots (attention).
- **Scope discipline:** Alex cautioned **not to go too deep on every facet** in the first empirical pass. The six mechanisms are **reasons the theory matters**; the paper should **address some, not all**, and use them to guide measurement—not prove six separate causal channels in one MBB table.

Capability vs. substitutability (critical PD12 fix)

The three-row table in `Alex_model.md` is easy to misread.

Misread (Charles, first pass)	Corrected read (Alex, PD12)
"Capability" = organizational strength of the program	"Capability" $\approx$ level and clarity of individual talent in the environment— how differentiated the person is from peers
Weak schools = low org quality	Weak environments = low absolute talent and low substitutability
Elite schools = high org quality only	Elite environments = very high talent but extreme substitutability —top individuals are excellent but hard to tell apart

Charles and Alex converged: in implementation, **capability** $\approx$ **latent** a_i (own performance, NBA composite, BPM, etc.), while **substitutability** is the new organizational object captured by **viable-peer congestion** $C_{i,t}$.

Implication for LOO constructs:

- **Congestion quality** (LOO mean of teammate `perf`, `poolq_loo`) $\approx \bar{a}_t$ / local peer caliber $\rightarrow$ can track the **rising** leg of an inverted-U (exposure, development, credibility).
- **Congestion crowding** (LOO count of teammates above viability threshold θ) $\approx C_{i,t}$ $\rightarrow$ should track the **downturn** (marginal candidates crowded out).
- At **fixed roster size**, LOO **sum** and LOO **mean** are almost the same information ($\sum \approx (n - 1) \times \text{mean}$); PD12 + Alex sim imply **do not treat sum as "crowding"**—use **threshold counts** (or smooth viability density as in `539_alex_model.ipynb`).

Alex on operationalizing crowding (~7:20): *"How many people are above the threshold—that's an easy calculation."* Then subtract a **small fraction** of that count from innate signal a_i in the evaluation equation (simulation); empirically, enter $C_{i,t}$ in $P(\text{draft})$ or promotion hazard.

Empirical hook discussed on the call: superstar tail of the draft curve

Charles connected the **small uptick at the very highest team-quality / pool-quality ventiles** in the 530 draft-rate plots to Alex's theoretical prediction that **elite superstars still get drafted** even when borderline players suffer from congestion.

- Alex had not yet mapped that plot to the theory; on PD12 he agreed: **"They still get up there in the very best teams."**
- **Use in writing:** The inverted-U is **not** "elite teams hurt everyone"—it is **heterogeneous**: top- a_i individuals clear the market; **middle-high** near-threshold players bear the congestion penalty. Heterogeneity tests in `Alex_model.md` are not optional flourishes—they are **the** discriminating prediction.

Army evidence shown to Alex (same meeting): 100-bin CIF, equal width

Charles showed Alex a **two-panel** figure from the Army pipeline ("**100 BINS, CA, EQUAL WIDTH**") to argue the downturn is **robust across systems**, not an NCAA-only curve-fit:

Panel	Outcome	Chart type
Top	Promotion (competing risks)	Final CIF by pool-quality bin (Q1–Q99)
Bottom	Attrition	Final CIF on the same binning / covariate axis

How to read the pair together:

- **Promotion (top):** Inverted-U-style pattern—gains from better pools up to a point, then a **downturn** in the highest bins (congestion / substitutability story for marginal promotable officers).
- **Attrition (bottom):** On the identical 100-bin axis, attrition is **not** a mirror image by construction (competing risks vs. separate cause-specific CIF). The **late-bin rebound** Charles called the **superstar tail** on this figure is visible on the **attrition** panel (CIF rises again toward Q70–Q99). Interpret with care: document whether top bins are **fewer people / wider uncertainty** before over-claiming; the intended PD12 point was **cross-domain shape exists**, not that promotion and attrition must move in opposite directions bin-for-bin.

Why Alex cared: One domain (MBB draft ventiles) can look like a house of cards; **Army promotion + attrition on the same pool construct** shows the broader VECTOR phenomenon is already in the tenure/Cox stack. That motivated confidence before investing in **congestion crowding** in MBB.

Figure file (workspace): user-provided Army CIF screenshot, May 2026 — promotion top, attrition bottom.

Priority workstreams (Alex’s agenda, second half of PD12)

Alex separated four threads. They are **not** the same task.

Priority 1 — Parameter identifiability (simulation + fit)

- Fit the generative model (~5–6 parameters) to **real data in all three domains** currently in scope (MBB, tenure, ...).
- Alex suspects the model is **overparameterized**: many parameter combinations may fit similarly.
- **Remedies discussed:**
 - **Fix** some generative knobs across worlds (e.g. same Beta hyperparameters everywhere—not estimated per dataset).
 - **Roll** mechanism-specific terms into a single outcome Y per domain (“in this dataset we roll a, b, c into Y ; in that one, a, b, d ”).
 - **Lower dimensionality** until recovered parameters are **unique and interpretable**, not just curve-fitting inverted-U shapes.

Distinction: Identifiability = “Can we estimate the structural parameters of *this* model from *these* tables?”

Priority 2 — Extreme events (simulation sensitivity)

- Not “rare draft surprises” in data—**knob-turning in the sim**: set $\lambda \rightarrow 0$, turn congestion off, tighten viability, etc.
- Confirm the inverted-U and mediation patterns **disappear mechanistically** when they should.
- Charles asked for clarification; Alex: validate the **internal logic** of `539_alex_model.ipynb` before over-interpreting one empirical ventile plot.

Priority 3 — Model-guided empirical features (different from Priority 1)

- The theory should **propose new measurements**, not only fit old ones.

- **Flagship example:** explicit **congestion** in MBB—quantify $C_{i,t}$, test whether it **mediates** the downturn in $\bar{a}_t$, Kentucky/Duke mega-recruit years as **quasi-experiments**.
- Charles (~12:15): ideal congestion measure **indirectly reflects** talent in the system but **does not enter the outcome as raw a_i** —so one can study “how crowded is this team?” and predict harm to marginal members. In practice θ is built from a_j or prospect signals (Y_{draft} , combine, recruiting); document that transparently.

Charles’s immediate commit after the call (~8:22): focus on adding the **congestion crowding** term in the empirical pipeline (not replacing all of 538, but making $C_{i,t}$ first-class).

Priority 4 — Falsification and scope (let seep over weeks)

- The three main systems were **chosen** because they plausibly satisfy the model’s conditions; a strong paper also shows **where it fails**.
- **Task:** Find a **fourth or fifth** setting where the inverted-U **does not** appear (NHL draft mentioned), and explain **which assumption breaks** (e.g. different attention structure, evaluation noise, no local peer congestion for slots).
- Also: **new positive domains** if data exist—Charles suggested **macro economics** (GDP, patents, oil production vs. GDP across nations competing for firms). Alex: “really nice level of estimation” at country–company scale.

Viable-peer signals in MBB data (connects PD12 to repo work)

`Alex_model.md` lists example thresholds for $1(a_j > \theta)$: top PPM share, draft probability, McDonald’s All-American, future pro, combine invitation.

Signal	Role in $C_{i,t}$	Repo status (May 2026)
<code>perf</code> / BPM / WS/40	Continuous a_j for θ or smooth v_j	<code>perf</code> on 530 panel; SR merge
LOO mean <code>poolq_loo</code>	$\bar{a}_t$ -like quality , not crowding	530, 538 CELL 10
LOO sum <code>pool_c_loo</code>	Near-redundant with mean at fixed n	538 crowding mode
Y_{draft}	Outcome + possible θ component	<code>athlete_id_draft_lookup</code>
Combine	Prospect visibility / θ	<code>combine_bridge.py</code> → <code>athlete_id_combine_lookup.csv</code> ; measurement row , not full invite list
Labeled union $Y_{\text{nba_signal}}$	“Serious NBA-facing peer” for thicker θ	Design discussed; not in panel export yet

PD12 did not discuss combine–draft union; separate design memo: union helps **peer identification**, not draft **outcome**; see bridge diagnostics (~58% of combine rows linked to `athlete_id`).

Natural experiments Alex named (empirical Priority 3)

- **Kentucky / Duke mega-recruiting years** — exogenous shocks to $C_{i,t}$ on a fixed roster.
 - **Transfer portal / NIL** — later waves of concentration (if timed in data).
 - **Cohort shock / DiD** — when a school adds several elite same-position players, **incumbent** near-threshold players' draft odds should fall **conditional on own a_i** .
 - **Mobility reversal** — leaving a congested elite program for a strong but less crowded team should **raise** advancement odds (harder in MBB single-season panel, easier in tenure).
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Suggested implementation order (repo)

1. `tier1_mechanism_vars.py` / 530 export: `peer_viable_count_loo` (and optional smooth $C_{i,t}$ from `539` logic); keep `poolq_loo` as quality leg.
 2. **538 CELL 10**: Third playground mode or replace "crowding = LOO sum" with **viable-peer count**; label UI clearly (quality vs. crowding).
 3. `539_alex_model.ipynb` : Cells for Priority 1–2 (fit / identifiability, parameter kill switches).
 4. `panel_rebuild` / `combine_bridge` : Optional `Y_combine_meas` , `Y_nba_signal` on `player_season_panel_530.csv` for θ robustness.
 5. **Writing**: Short PD12 cross-ref in advisor briefs; do **not** claim all six mechanisms are separately identified in MBB v1.
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One-paragraph paper sentence (PD12-aligned)

We distinguish **team quality** (mean peer performance, $\bar{a}_t$) from **viable-peer congestion** (density of teammates above a prospect threshold, $C_{i,t}$). Elite environments raise both average talent and **substitutability**; the inverted-U in advancement is predicted to steepen on the right primarily where $C_{i,t}$ is high, especially for **near-threshold** individuals, while **top latent talent** still clears selection—consistent with the small recovery at the highest ventiles in our NCAA draft data.

Open questions left on the table

- Exact θ for MBB: global PPM quantile vs. combine/draft-based Y_{nba_signal} vs. smooth logistic viability (539).
- Whether to **mediate** with $C_{i,t}$ in LPM only or also in Tier 1 generative selection (538).
- NHL (or other) **negative case** data feasibility for Priority 4.
- Country-level economics as fourth **positive** domain vs. falsification case.

Addendum last updated to reflect PD12 transcript and repo state as of May 2026.